

Prime neighbours of zeta denominators

Zeta doors: a computational note

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Abstract

For positive even indices k with $3 \nmid k$, let D_k be the reduced positive denominator of $\zeta(1 - k)$. Classical Bernoulli arithmetic gives $D_k = 3 \cdot 2^{1+v_2(k)} T_k$, where every prime factor of T_k is 2 modulo 3. If also $4 \nmid k$, these factors are 11 modulo 12; otherwise factors congruent to 5 modulo 12 are also permitted. We select the unique neighbour of $D_k/3$ that is not divisible by 3 and investigate when it is prime. The construction provides known factorisations on one side of each candidate and therefore supports classical primality certificates. We give the elementary selection rule, describe its limited Lean formalisation, and record computational examples and the verification conditions needed for their interpretation. No claim of an infinite prime family or of priority for the construction is made.

1 The observation and the construction

The equality

$$\zeta(-1) = \zeta(-13) = -\frac{1}{12}$$

motivated an examination of arguments $1 - k$ with $k \equiv 2 \pmod{12}$. The equality itself does not establish a period of the zeta function. Our object is the denominator at these arguments, rather than equality of their values. The construction extends to all positive even indices not divisible by 3, namely $k \equiv 2, 4, 8, 10 \pmod{12}$. We call $k \equiv 2 \pmod{12}$ the original family and the other three residue classes the added family.

For a rational number x , write $\text{den}(x)$ for its positive denominator in lowest terms. For a prime p , let $v_p(k)$ denote the exponent of p in the positive integer k . Throughout the note,

$$k \geq 2, \quad 2 \mid k, \quad 3 \nmid k, \quad D_k = \text{den}(\zeta(1 - k)).$$

For an integer n not divisible by 3, put

$$\chi_3(n) = \begin{cases} 1 & n \equiv 1 \pmod{3}, \\ -1 & n \equiv 2 \pmod{3}. \end{cases}$$

For an odd prime $p \neq 3$, its *3-free door* is $m_0(p) = p + \chi_3(p)$: the even neighbour not divisible by 3.

2 Classical denominator arithmetic

With the convention $B_1 = -1/2$, the Bernoulli numbers satisfy $\zeta(1 - k) = -B_k/k$ for positive even k . The classical denominator formula for these divided Bernoulli numbers is

$$D_k = \prod_{\substack{p \text{ prime} \\ p-1 \mid k}} p^{1+v_p(k)}. \tag{1}$$

For completeness, von Staudt–Clausen gives $v_p(B_k) = -1$ when $p - 1 \mid k$. Hence $v_p(B_k/k) = -1 - v_p(k)$ for those primes. For the other primes, the classical integrality property of divided Bernoulli numbers gives $B_k/k \in \mathbb{Z}_p$; see Kellner’s account [1]. Together these facts give (1). In particular, dividing the denominator of B_k by k without accounting for cancellation would not justify this formula.

Proposition 1 (Extended selection rule). *If $p > 3$ is prime and $p - 1 \mid k$, then*

$$p \equiv 2 \pmod{3}, \quad m_0(p) = p - 1 \mid k.$$

Consequently $p \equiv 5$ or $11 \pmod{12}$. If in addition $4 \nmid k$, then $p \equiv 11 \pmod{12}$.

Proof. If $3 \mid p - 1$, then $p - 1 \mid k$ would imply $3 \mid k$. Thus $p \not\equiv 1 \pmod{3}$. Primality and $p > 3$ exclude $p \equiv 0 \pmod{3}$, leaving $p \equiv 2 \pmod{3}$. This gives $m_0(p) = p - 1$. Since p is odd, its possible residues modulo 12 are 5 and 11. If $4 \nmid k$, then $4 \nmid p - 1$; oddness now forces $p \equiv 3 \pmod{4}$, selecting 11 modulo 12. $\square$

The stronger conclusion therefore covers both $k \equiv 2$ and $k \equiv 10 \pmod{12}$. For $k \equiv 4$ or $8 \pmod{12}$, factors congruent to 5 are permitted, not required.

Corollary 2. *Put $a_k = 1 + v_2(k)$ and*

$$T_k = \prod_{\substack{p > 3 \text{ prime} \\ p-1 \mid k}} p^{1+v_p(k)}.$$

Then

$$D_k = 3 \cdot 2^{a_k} T_k, \quad d_k = D_k/3 = 2^{a_k} T_k.$$

The integer T_k is odd and not divisible by 3; all its prime factors are 2 modulo 3. Thus d_k is even and not divisible by 3. When $4 \nmid k$, one has $a_k = 2$, recovering $D_k = 12T_k$.

Proof. In (1), the prime 2 occurs to exponent $1 + v_2(k)$, and 3 occurs to exponent 1 since $v_3(k) = 0$. The remaining factors obey Proposition 1. $\square$

Definition 3. *The zeta-door candidate at k is*

$$Q_k = d_k + \chi_3(d_k).$$

We call it a zeta-door prime when Q_k is prime.

This selects exactly one of $d_k - 1$ and $d_k + 1$: the other is divisible by 3. If Q_k is prime, its residue modulo 3 is opposite to that of d_k , so

$$m_0(Q_k) = d_k. \tag{2}$$

Writing $\Omega(T)$ for the number of prime factors counted with multiplicity, including $\Omega(1) = 0$, an equivalent expression is

$$Q_k = 2^{a_k} T_k + (-1)^{a_k + \Omega(T_k)}.$$

The exponent multiplicities and the parity of a_k are essential. For $4 \nmid k$ this reduces to $Q_k = 4T_k + (-1)^{\Omega(T_k)}$.

3 Small examples and the relation to doors

Direct rational evaluation gives the following examples.

k	$1 - k$	D_k	d_k	Q_k
2	-1	12	4	5
4	-3	120	40	41
8	-7	240	80	79
10	-9	132	44	43
14	-13	12	4	5
26	-25	12	4	5
38	-37	12	4	5
50	-49	132	44	43

Thus different indices can produce the same candidate. Counts of indices, distinct denominators, and distinct primes must be separated. The table concerns reduced denominators; it does not assert that the zeta values with the same denominator are equal. For $k = 8$ and 10, the eligible neighbours are on the minus side: $79 = 80 - 1$ and $43 = 44 - 1$. Their known doors factor $Q_k + 1$, not $Q_k - 1$.

Two divisibility relations also require separation. The denominator selection rule reads $m_0(p) \mid k$ for a contributing prime $p > 3$. The ownership relation in the hire construction reads $p \mid m_0(Q)$. When $Q = Q_k$ is prime, (2) implies that every prime factor of T_k is an owner-door factor of Q . These observations do not identify the index k with a graph vertex or establish a theorem about finite gold components or Laplacian spectra.

4 Search and certification

The search starts with a fully factored index k . It enumerates all positive divisors $d \mid k$ and examines $p = d + 1$ for primality. Equation (1) then constructs D_k without computing the enormous numerator of B_k/k . Small-factor sieving and probable-prime tests screen the resulting Q_k . Screening is distinct from proof.

The targeted runs recorded here use the plus branch $Q_k = d_k + 1$. Consequently $Q_k - 1 = d_k$ has a supplied factorisation. If $N = Q_k$ and $N - 1 = FR$, where F is completely factored into proved primes and $F^2 > N$, a partial Pocklington certificate suffices. For each prime $r \mid F$, the certificate supplies a witness a_r with

$$a_r^{N-1} \equiv 1 \pmod{N}, \quad \gcd(a_r^{(N-1)/r} - 1, N) = 1.$$

These conditions force every prime divisor of N to be 1 modulo F ; since $F > \sqrt{N}$, N must be prime. This standard criterion is the primality engine, not a new primality theorem; see [2]. Minus-branch candidates would instead expose a factorisation of $Q_k + 1$ and need a suitable different certification method.

Two independent obligations remain:

1. Prove the candidate Q_k prime. Only enough proved factors of $Q_k - 1$ to meet Pocklington's bound are needed.
2. Prove the claimed denominator is exact. Every included $d + 1$ must be prime, and every omitted $d + 1$ must be composite.

For a large contributing prime p , the factorisation of $p - 1 \mid k$ is already available from that of k . A complete $p - 1$ certificate can therefore establish its primality at relatively low cost.

The verifier uses a deterministic Miller–Rabin test in the range $n < 2^{64}$ and Pocklington above that range. Its seven Miller–Rabin bases are

$$2, 325, 9375, 28178, 450775, 9780504, 1795265022.$$

The range restriction is part of the verification contract.

5 A finite comparison of the two families

The committed census at $k \leq 10000$ compares the original residue class with the added classes. Denominators and candidate values are deduplicated separately within each column.

	$k \equiv 2 \pmod{12}$	$k \equiv 4, 8, 10 \pmod{12}$
Indices	834	2500
Distinct D_k	171	1426
Repeated-denominator occurrences	663	1074
Distinct plus-side screen survivors	29	160
Distinct eligible screen survivors	44	320

The source is `family-census-10k.json`, generated by `compare_families.py` in the targeted-search directory. Repeated means every occurrence after the first within its family. About 79.5% of original-family indices repeat a denominator, compared with 43.0% in the added family. The added family contains about three times as many indices, so raw prime counts alone are not a comparison of search efficiency. The columns can share denominators and primes; their distinct counts must not be added to obtain a union count.

The census calls the search engine’s primality predicate, rather than producing a certificate for every survivor. Its stored keys use the word primes; the table deliberately records them as screen survivors until deterministic checks or certificates establish their status. In particular, the plus-side certificate workflow does not certify minus-side survivors through their known door. The census is separate from the completed large candidate certificates below.

6 Computational snapshot and reproducibility

The notebook reports the following successful candidate-primality checks in the original-family `kitchen8` run. These are experimental records of this search, not claims of global size records.

Digits of Q_k	Distinct primes below 2^{64} in F	Larger listed factors
5522	510	5
5118	462	3
4443	360	7

For the 5522-digit entry the index is

$$k = 111695474196999960960050.$$

The author reports that all 5, 3, and 7 larger listed factors, respectively, were subsequently proved prime from their index factorisations. Three plus-side candidates from the same `lions71` hunt are certified and independently verified (Mac Studio, Apple M2 Max, 12 cores, 32 GB):

Digits of Q_k	<code>cert.py</code>	F primes $< 2^{64}$	Cofactors	<code>verify.py</code>
18365	9 h 2 min	1598	19	24 Sep 2026
16213	5 h 19 min	1215	58	25 Sep 2026
16165	5 h 19 min	1221	62	25 Sep 2026

The 18365-digit index is $k = 1522610689958516981964050$ (`lions71/hits.certified.jsonl`). The two smaller rows are `lions71/hits.smaller.certified.jsonl`. In each case $F^2 > N$ from the 64-bit part, and listed cofactors were proved from the factored index.

The source snapshot for this draft is 25 September 2026 [3]. Change into the directory

`analysis/zeta/zeta-targeted`

and run

```
python3 verify.py kitchen8/hits.jsonl
python3 verify.py lions71/hits.certified.jsonl
python3 verify.py lions71/hits.smaller.certified.jsonl
```

Use ordinary Python execution, without optimisation flags that disable assertions. The JSON lines contain the exact candidates, indices, factorisations, and witnesses; they are preferable to printing thousands of decimal digits in the note. All three `lions71` certificates have been independently rechecked as above.

Outstanding audit item for this draft. Revision `c04d373` searches bases $2, \dots, 501$ when proving large divisor-derived candidates. Exhausting this bounded search returns `False`, which the denominator loop treats as a reason to omit the candidate. Failure to find a primality witness is not a compositeness proof. A publication-ready exact-origin receipt must distinguish *prime*, *composite*, and *unresolved*, and must not finish successfully with unresolved divisor candidates. The successful proofs of all included large factors do not by themselves settle this exhaustiveness issue. This qualification concerns denominator provenance, not the separate partial-Pocklington proof of Q_k .

7 Formalisation and further questions

The file `lean/Hire/ZetaDoors.lean` now contains the following theorems (names are given exactly for navigation):

- `prime_mod_three_of_pred_dvd_even`: the extended conclusion $p \equiv 2 \pmod{3}$.
- `m0_dvd_of_pred_dvd_even`: the consequence $m_0(p) \mid k$.
- `prime_mod_twelve_of_pred_dvd_not_four`: the sharper residue 11 modulo 12 under $4 \nmid k$.
- `prime_mod_twelve_of_pred_dvd_four`: the alternatives 5 or 11 modulo 12. Despite its name, its statement does not require $4 \mid k$; it applies to the full extended family.

The original `prime_mod_twelve_of_pred_dvd_index` and `m0_dvd_of_pred_dvd_index` remain for $k \equiv 2 \pmod{12}$. The residue-class lemma `even_not_three_mod_twelve` identifies the four admissible index classes. The source mentions Mathlib’s von Staudt–Clausen theorem but does not invoke it. It does not formalise (1), the complete search, or the large numerical primality certificates. The Lean source was inspected for this update; its compilation was not independently repeated.

The immediate computational questions are how often denominators repeat, how the two signs are distributed, and how prime frequency changes under different index-selection schemes. A density measured in a targeted sample of highly factored indices should not be presented as the density over all even k with $3 \nmid k$. Whether infinitely many distinct zeta-door primes exist is not settled here. A literature comparison is also needed before making a priority claim for this particular construction.

The companion Lab at <https://halfasecond.com/> retains changing records and experiments; this note fixes definitions, proof scope, and a dated computational snapshot.

References

- [1] B. C. Kellner, *The Bernoulli Number Page*, sections on denominator and numerator structure, <https://www.bernoulli.org/>. Accessed 22 September 2026.
- [2] PrimePages, *Finding primes and proving primality*, Section 3: classical primality tests, <https://t5k.org/prove/prove3.html>.
- [3] J. Pickett, *Natural Maths notebook*, source and computational records, revision c04d373, <https://github.com/hasjack/nm-notebook/tree/c04d373>.